Machine Learning (2026) 115:252
https://doi.org/10.1007/s10994-026-07185-w

Optimizing AlphaPose with Dual Non-Local Attention for Real-Time Unsafe Behavior Detection in Mining Environments

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Received: 8 May 2026 / Revised: 4 August 2026 / Accepted: 30 September 2026
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Abstract
In the mining industry, unsafe worker behaviors pose significant safety risks. Recent advances in deep learning have improved unsafe behavior recognition; however, challenges remain in achieving both real-time performance and high accuracy under complex mining conditions, as well as in the availability of dedicated datasets. To address these issues, we introduce a comprehensive miner unsafe behavior dataset and propose the Mining Industry Unsafe Behaviors Monitoring Framework (MIUBM-Net), which integrates STFNet-A for typical unsafe behavior detection and Alpha-SDL-Pose, based on Dual Non-Local Attention (DNLA), for posture-related unsafe behavior recognition. STFNet-A achieves 96.9% mAP at 59.6 FPS, while Alpha-SDL-Pose improves posture recognition accuracy to 94.9% at 49.4 FPS. Experimental results demonstrate the effectiveness of the proposed framework and its constituent modules for recognizing typical and posture-related unsafe behaviors in underground mining environments, providing a practical solution for intelligent mining safety monitoring.

Keywords Intelligent mining · Unsafe behaviors recognition · Attention mechanism · Object detection · Pose estimation

Editor: Mingming Gong.

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Published online: 26 October 2026

1 Introduction

Currently, many countries are enhancing the supervision of production safety in coal mining enterprises. Many safety regulations have been revised to govern safety production in coal mining enterprises and ensure the personal safety of coal miners. Statistical analyses indicate that a considerable number of severe accidents in the coal mining industry are attributed to miners' unsafe behaviors (Yang et al., 2022; Niu et al., 2023). However, most accidents can be avoided by regulatory measures, so monitoring, identification and early warning of miners' unsafe behaviors are essential to prevent accidents.

Deep learning is a neural network-based machine learning method that plays a crucial role in artificial intelligence. In computer vision, deep learning has demonstrated remarkable performance. It has achieved milestone breakthroughs in tasks such as image processing (Chakraborty & Mali, 2023; Suganyadevi et al., 2022), target detection (Zhao, 2022; Wen et al., 2023), pose estimation (Liu et al., 2023; Yang et al., 2023), and image segmentation (Cheng et al., 2023; Yang & Cai, 2023). With artificial intelligence technology's continuous development and improvement, mine construction has changed from the traditional information mode to the intelligent mode. The study of miners' unsafe behavior recognition has become popular among many experts and scholars (Jiang et al., 2022; Roy & Bhaduri, 2023; Roy et al., 2022).

The single-stage target detection algorithm YOLO series, SSD algorithm and two-stage target detection algorithm RCNN series have achieved good results in detecting unsafe behaviors such as helmets and masks (Mathur & Jain, 2023; Umer et al., 2023; Lee et al., 2023). Unlike other methods that combine Transformer and YOLO models, the proposed STFNet-A is specifically designed for the mining industry. STFNet-A utilizes the Swin Transformer as the backbone and constructs a feature pyramid. Additionally, the algorithm incorporates Quality Focal Loss and the CBAM (Convolutional Block Attention Module) attention mechanism to address common issues of sample distribution imbalance and scene noise in mining environments, thereby adapting to the unique needs of the mining field.

Newell et al. (2016) designed Stacked Hourglass Network for human pose estimation. The network uses three strategies, Bottom-Up, Top-Down, and Intermediate Supervision, to improve network performance and predict human keypoints by generating a set of heat maps. The AlphaPose system (Fang et al., 2017), developed by Cewu Lu's team at Shanghai Jiao Tong University, effectively detects multiple targets and adopts the Regional Multi-Person Pose Estimation (RMPE) framework to achieve accurate human pose estimation, providing an effective solution for posture-related unsafe behavior recognition.

However, existing human pose estimation and action recognition methods are mainly developed for general scenarios and cannot fully satisfy the requirements of safety monitoring in underground mining environments. Although existing human action recognition datasets contain various behaviors such as sitting, crossing, and kicking, these datasets are mainly collected from daily-life scenarios and are not specifically designed for safety monitoring applications in underground mines. The visual characteristics of miners' unsafe behaviors differ significantly from ordinary human actions due to the complexity of mining environments. Factors such as low illumination, heavy equipment occlusion, protective equipment interference, dust, and irregular surveillance viewpoints may cause significant variations in human appearance and motion patterns. Therefore, directly applying general action datasets may not adequately represent the safety-critical scenarios encountered in

underground mining, highlighting the necessity of constructing a dedicated miner unsafe behavior dataset. A comparison between existing human action datasets and the proposed miner unsafe behavior dataset is summarized in Table 1.

Recognizing unsafe behaviors is an advanced and complex visual perception task aimed at understanding and evaluating human activities. To address the above challenges, we have done the following work based on analyzing unsafe behavior recognition (Yu et al., 2023; Chen, 2022).

(1) To address the limitation of lacking a dedicated dataset for mining safety monitoring, the unsafe behaviors of miners were systematically analyzed, and a miner unsafe behavior dataset was constructed for safety monitoring model training. Different from existing human action datasets that mainly focus on general activity recognition, the proposed dataset was developed according to coal mine safety risk database standards and considers the characteristics of underground mining scenarios. The collected behaviors were divided into two categories: typical unsafe behaviors and posture-related unsafe behaviors. The typical unsafe behaviors include six categories: correctly wearing a helmet (hat), incorrectly wearing a helmet (err-hat), not wearing a helmet (no-hat), correctly wearing a mask (mask), incorrectly wearing a mask (err-mask), and not wearing a mask (no-mask). The posture-related unsafe behaviors include illegal sitting and lying, illegal kicking and punching, and illegal crossing. Furthermore, various data augmentation strategies were employed to improve model robustness and adaptability under diverse mining conditions.
(2) A target detection model STFNet-A is proposed to achieve real-time and accurate detection of miners' routine unsafe behaviors in complex operating environments of coal mines. Firstly, we use Swin Transformer as the backbone network and build a feature pyramid, design the target detection model architecture STFNet, introduce the Quality Focal Loss function(QFL) to focus on the intra-class differences of samples, and effectively solve the problem of the unbalanced sample distribution. Then, to further suppress the scene noise problem, CBAM attention is introduced to fuse local and global contextual information to build model STFNet-A, which achieves 96.9% accuracy for typical unsafe behaviors recognition, 59.6FPS detection speed, and model parametric number of 31.3M.
(3) An optimized model based on AlphaPose, Alpha-SDL-Pose, is proposed to achieve miners' posture-related unsafe behaviors detection in complex operating environments of coal mines. Firstly, we use STFNet-A to replace AlphaPose's Faster R-CNN to improve the detection speed and reduce the number of model parameters. Then, we design the DNLA module and embed the pose estimator to capture longer distance and richer contextual information to extract feature information such as the target person's

Table 1 Comparison between general action datasets and the proposed miner unsafe behavior dataset

Dataset	Scenario	Lighting	Occlusion	Safety Behaviors
General action	Daily-life	Normal	Limited	No
Pose datasets	General human	Various	Limited	No
Indoor action	Indoor env.	Normal	Moderate	No
Proposed dataset	Under-ground mine	Low light	Heavy	Yes

joint point location, joint linkage and skeletal pose. Finally, the dangerous actions are accurately classified based on LSTM unsafe behavior classification network. The model improves the recognition accuracy by 2.7 percentage points over AlphaPose to 74.8% and increases the detection speed from OpenPose's 18.6 FPS to 49.4 FPS on the MSCOCO 2017 public dataset of over 250,000 human pose samples. The recognition accuracy of miners' posture-related unsafe behaviors on the homemade dataset reached 94.9%, and the model parameters were 43.9M.

Experimental results demonstrate that STFNet-A and Alpha-SDL-Pose achieve favorable trade-offs between accuracy and efficiency for typical and posture-related unsafe behaviour recognition, respectively, providing an effective framework for intelligent mining safety monitoring.

2 Related Work

Deep learning-based unsafe behavior recognition is a fundamental and challenging task in computer vision, which is widely used in intelligent security (Mahor et al., 2023; Bhardwaj et al., 2023), industrial production (Rafiei et al., 2023; Aparicio Pico et al., 2023), autonomous driving (Bendiab et al., 2023; Wang et al., 2023), competent healthcare (Balasundaram et al., 2023; Elango et al., 2023), and smart cities (Verhulsdonck et al., 2023; Ahmed et al., 2023). These approaches have substantially improved by combining the rich information of context (Park et al., 2020) and structured prediction techniques (Shenoy et al., 2020; Liang et al., 2023).

In contrast, recognizing unsafe behaviors of miners in underground coal mines is critical, and research from the perspective of computer vision and deep learning is still in its infancy. Qiao and Chen (2022) defined the concept of big data in mines. They proposed three operational models for predicting miners' dangerous behaviors. However, the cost of these models is relatively high due to the limitations of big data processing techniques and a large amount of data high.

Kim et al. (2021) successfully addressed the helmet-wearing problem using various deep-learning techniques and significantly improved. However, this approach was limited to detecting miners' helmets and could not effectively identify other potential risk factors. Using convolutional neural networks, Xiang et al. (2022) developed a model to evaluate whether miners wear helmets. To effectively prevent personnel from trespassing into hazardous areas and reduce misjudgments of unsafe behaviors, precise positioning of underground workers is essential. Currently, positioning techniques are mainly based on wearable devices such as smart helmets (Mukhopadhyaya et al., 2022), smart bracelets (Li, 2022), and mining watches (Agboola et al., 2020) to improve safety and portability. However, long-term wear may reduce efficiency and lead to devise damage, making it difficult to cope with emergencies.

Dai-Hong et al. (2019) used a surveillance video target tracking method to unfold the unsafe underground miners. Although this method can be applied in darker environments, its recognition accuracy still needs further improvement. Chen and Li (2021) proposed a motion target technique for monitoring underground, which can perform tracking of motion

targets, but lacks assessment judgment of miners' unsafe behavior. Lu et al. (2020) proposed a technique to monitor the operational status of miners to identify facial expressions for monitoring operators. Wu et al. (2023) proposed a re-identification method for underground personnel, which can effectively track and monitor the behavior of personnel working in coal mines to ensure safety.

Singh et al. (2022) proposed a new method that organically integrates data and relevant knowledge to identify miners' unsafe behaviors better. They also predicted the potential of artificial intelligence in mining IoT. You et al. (2023) used migration learning techniques to construct a new residual network model to detect miners' unsafe behaviors effectively. However, this model still requires expensive and time-consuming image processors. Wang et al. (2022) proposed a computer vision-based unsafe behavior recognition system that can effectively collect, store, and analyze underground surveillance videos and can achieve accurate identification of unsafe behaviors, effective model management, and early warning triggering warning management.

In general, the underground coal mine is a complex and changing production environment. To ensure miners' safety, it is essential to make judgments based on the working environment and provide timely warnings to effectively prevent accidents (Zhang et al., 2022). Working in a restricted space, miners are prone to fluke, paralysis and slackness, and these psychological factors can impact their behavior, leading to conscious or unconscious unsafe behaviors. In addition, some miners are prone to unsafe behaviors by ignoring regulatory rules, kicking and punching equipment in the workplace, and sitting and resting illegally. These research results demonstrate the potential of computer vision and deep learning techniques for miners' unsafe behavior recognition. However, more research is still needed to address the technical limitations and practical applications.

3 Methodology

3.1 STFNet-A for Typical Unsafe Behaviors

3.1.1 STFNet-A Overview

Observation and subjectivity play vital roles in human practice. We propose a lightweight target detection algorithm STFNet-A, drawing on the YOLOV5 structure, which aims to provide better support for routine unsafe behavior detection.

Its structure diagram is shown in Fig. 1,The network consists of three main components: the Swin Transformer-based backbone, the feature fusion neck, and the detection head. The backbone extracts hierarchical feature representations from input images, while the neck performs multi-scale feature fusion through convolution and upsampling operations. The detection head further enhances discriminative features using the CBAM attention module for typical unsafe behavior recognition.

(1) Input: The image input module takes a series of data pre-processing techniques for the input images.

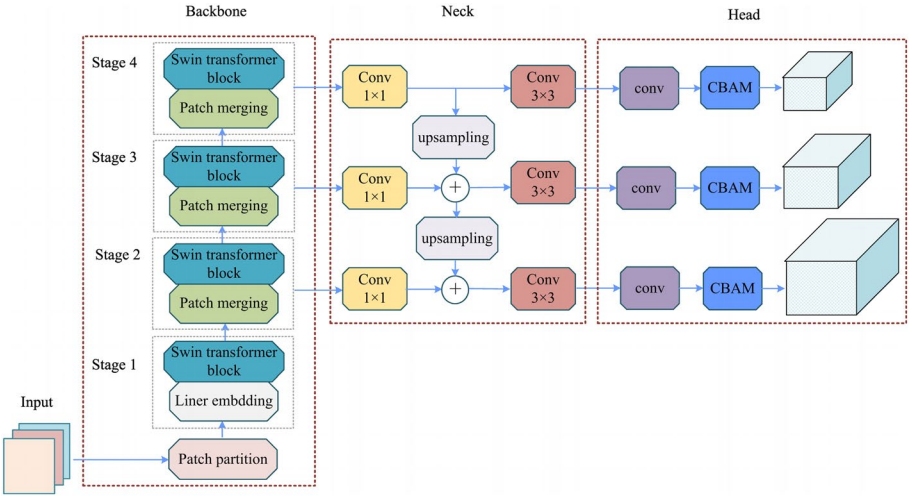


Fig. 1 STFNet-A network structure

(2) Backbone: Swin Transformer is the backbone network to extract the image features. The feature map is downsampled through 4 stages, each time with 2x downsampling, 2x height and width reduction and doubling the number of channels.
(3) Neck: The last three layers of the feature extractor are used to build the feature pyramid, and each layer is used to predict the detection frame. The feature pyramid has two paths, from the top down and from left to right. The path from left to right adjusts the number of channels of the feature map by 1×1 convolution, and the path from the top down upsamples the higher-level features, adds and fuses the higher-level features with their neighbouring lower-level features, and enhances the semantic information of the lower-level feature map.
(4) Head: The detection network includes localization and classification networks. The module introduces CBAM attention to further suppress the scene noise problem and fuse local and global contextual information. Because of the extremely uneven distribution of dataset samples in the coal mine scenario, Quality Focal Loss is used to train the network. This reduces the relative loss of easily classified samples and focuses the network on complex and misclassified samples.

3.1.2 Feature Extraction: Swin Transformer Backbone

The feature extraction module (Swin Transformer) contains patch partition layer, patch expanding layer, patch merging layer and swin transformer layer, where patch expanding and two swin transformer layers form phase one. The feature extraction layer, patch merging layer and multiple swin transformer layers form the second, third and fourth feature extraction stages. The model first slices the input image with height H, width W, and the number of channels 3, and the slicing result is $\frac{H}{4} \times \frac{H}{4} \times 48$. Then after four stages of feature extraction, three feature maps are finally output, and their dimensions are $\frac{H}{8} \times \frac{H}{8} \times 2C$, $\frac{H}{16} \times \frac{H}{16} \times 4C$,

and $\frac{H}{32} \times \frac{H}{32} \times 8C$, where C is the number of channels. At the patch partition layer, the feature maps are sliced with a 20% reduction in resolution while the lost pixel information is added to the channels. The patch expanding layer keeps the height and width of the feature map unchanged and only maps the number of channels to C. The patch merging layer reshapes the feature map by reducing its height and width by half while increasing the channel dimension by a factor of two. The swin transformer layer is used for feature learning and does not change the size of the feature map. Each stage needs to be used in pairs swin transformer layers used for feature learning and do not change the size of the feature map, and each stage requires pairs of swin transformer layers, and the number of layers must be even. Each swin transformer layer consists of layer normalization, multi-headed self-attention mechanism, residual structure, and multilayer perceptron, and the structure is shown in Fig. 2.

The block consists of two successive transformer layers, including Window-based Multi-head Self-Attention (W-MSA) and Shifted Window-based Multi-head Self-Attention (SW-MSA), followed by Layer Normalization and Multilayer Perceptron (MLP) modules. The self-attention mechanism models relationships between image patches by generating query, key, and value representations.

These layers work together to achieve feature learning and processing. The multi-headed self-attention module can be described as different self-attention mechanisms applied in two consecutive layers. The first layer uses window-based multi-head self-attention (WMSA), and the second uses shifted window-based multi-head self-attention (SWMSA). The relevant formulas are given by Eqs. (1)-(4).

$$\hat{z}^l = WMSA\left(LN\left(z^{l-1}\right)\right) + z^{l-1} \tag{1}$$

$$z^l = MLP\left(LN\left(\hat{z}^l\right)\right) + \hat{z}^l \tag{2}$$

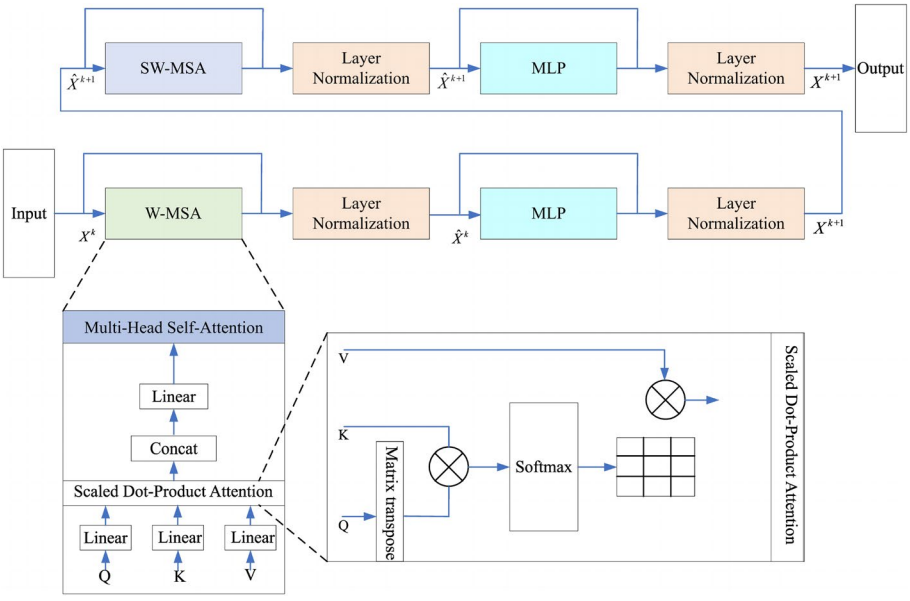


Fig. 2 Swin Transformer layer structure

$$\hat{z}^{l+1} = SWMSA\left(LN\left(z^{l}\right)\right) + z^{l} \tag{3}$$

$$z^{l+1} = MLP\left(LN\left(\hat{z}^{l+1}\right)\right) + \hat{z}^{l+1} \tag{4}$$

Where z^{l-1} denotes the output features of layer $l-1$ MLP, $\hat{z}^{l}$ denotes the output features of layer l WMSA, z^{l} denotes the output features of layer l MLP, $\hat{z}^{l+1}$ denotes the output features of layer $l+1$ SWMSA, and z^{l+1} denotes the output features of layer $l+1$ MLP.

The swin transformer layer uses two improved window-based self-attention modules: the W-MSA module and the SW-MSA module. These modules can effectively reduce the computation by putting the attention computation into the segmented window. Combining the two modules can better fuse the global features and improve detection efficiency. In addition, the Swin Transformer layer uses a hierarchical feature representation to capture features at different scales. This is achieved by adding multiple levels of Transformer encoders to the network, with the lower level encoder responsible for extracting local and the higher level encoder responsible for extracting global features. This hierarchical feature representation can better capture the image. This layered feature representation can better capture the multi-scale information in the image. The hierarchical feature map and the panning window method are shown in Fig. 3.

3.1.3 Feature Compression and Fusion: Neck Module

The neck network (Neck) is a structure that connects the feature extraction module (Backbone) and the detection network (Head) in deep learning for feature compression and fusion. Since the feature maps extracted by the feature extraction module are on the large side, but

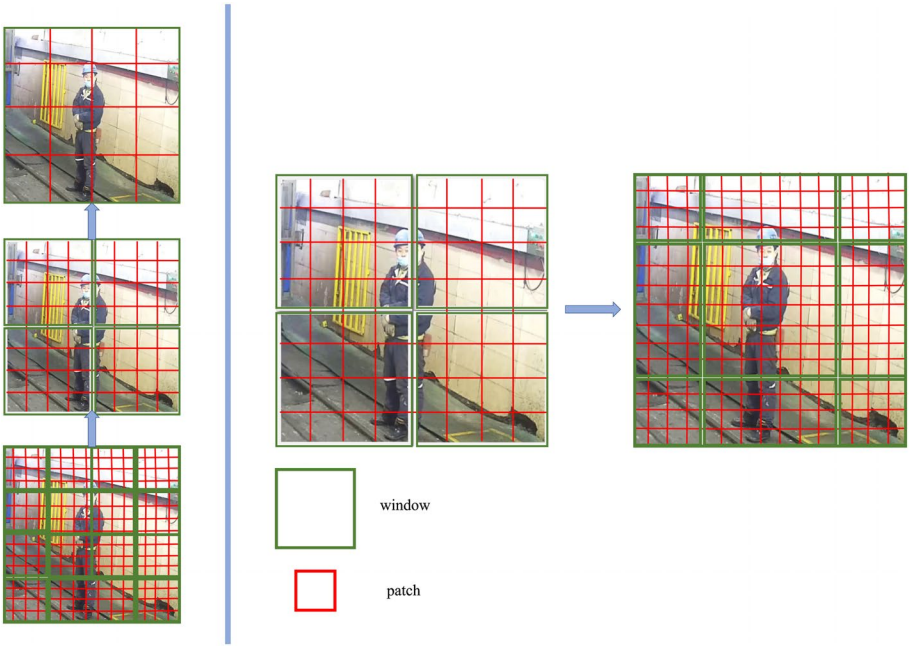


Fig. 3 Hierarchical feature map and shifted window approach

the dimensionality of the output is relatively small, the Neck structure is designed to further process these features for better transfer to the detection network for predicting the results. In this paper, the feature pyramid is constructed using the feature maps outputted from the last three stages of the swin transformer. The lower tiers of the feature pyramid are rich in fine-grained spatial details, while the high-level features contain rich semantic information. In order to make full use of the low-level and high-level features and improve the detection accuracy of typical unsafe behaviors, the feature pyramid fuses the high-level features and the low-level features. The fused features require adjusting the number of channels of the feature map by a 1×1 convolution kernel and upsampling the high-level features so that the dimensions of the high-level features and their adjacent low-level features are consistent and then summing them to obtain the fused features.

3.1.4 Multi-scale Target Classification and Localization: Head Module

The Head module implements multi-scale target classification and localization. The output of the Neck module contains rich, detailed features and semantic information, and we embed CBAM attention in the Head module to filter and analyze the input feature information in the channel dimension and spatial dimension, respectively, to enhance the impact of essential features and thus obtain performance gains.

3.1.5 Loss Function

Focal Loss is an effective method to solve the sample imbalance problem, especially for tasks with extremely unbalanced positive and negative sample ratios, such as identifying typical unsafe behaviors of miners in this topic. In addition, it has anti-noise and anti-spoofing properties, which can help the model better handle noise and anomalies. However, a significant drawback of Focal Loss (FL) is that there may be a significant bias between training and inference. Specifically, the scores of learning categories and localization are computed separately in training. However, the product of the two is used in inference to perform non-maximal suppression operations, which may lead to low confidence in some detection frames and thus affect the final detection accuracy.

The samples must be wearing masks correctly, and not wearing masks in the typical unsafe behavior dataset accounts for a relatively small overall sample. The positive and negative samples could be more balanced during the training period. In order to effectively solve the above problems, this paper adopts a combination of classification score and localization IoU to unify the learning category score and localization score. The improved expression of Quality Focal Loss (QFL) is shown in Eq. (5).

$$QFL(\sigma) = -|y - \sigma|^{\beta}((1 - y)\log(1 - \sigma) + y\log(\sigma)) \tag{5}$$

Where y denotes the quality labels from 0 to 1, and σ denotes the predicted results.

3.1.6 Datasets

We use a cooperative coal mine video surveillance dataset that intercepts eligible images containing various scenes of coal miners at work. Various data enhancement techniques

such as HSV transform (Smith, 1978), scaling and cropping (Avidan & Shamir, 2007), and CutMix fusion (Yun et al., 2019) is introduced to enhance the robustness, generalization ability, and adaptability to different data distributions of the deep learning model. A total of 11,600 images from the typical unsafe behaviors detection sample library are used to train and validate the model to detect different unsafe behaviors. To accommodate different detection needs, the dataset was made into six categories, including correctly wearing a hat (hat), not wearing a hat correctly (err-hat), not wearing a hat (no-hat), correctly wearing a mask (mask), not wearing mask correctly (err-mask), and not wearing a mask (no-mask). To train and evaluate the model, we randomly divide the dataset into training and validation sets, where the training set accounts for 80% and the validation set accounts for 20%.

3.1.7 Software and Hardware Configuration

The experiments in this study were conducted in a software environment comprising Ubuntu, PyTorch 2.0, Python 3.9.13, CUDA 11.8, and OpenCV 4.9.0.80. The hardware setup includes an Intel Core i7-12700K processor and an NVIDIA RTX 3060 graphics card. A consistent software and hardware environment was maintained across all experiments detailed in Sect. 3.2.

3.1.8 Experiment

3.1.8.1 Performance Comparison of Different Methods In order to verify the effectiveness, accuracy and feasibility of the proposed algorithm in this paper, we conducted several experiments on the conventional unsafe behavior dataset and performed experimental analysis. As mentioned above, the accuracy and efficiency of miners' unsafe behaviors identification in extreme environments are important indicators of the method. Therefore, we evaluate the model's performance by two quantitative metrics, accuracy and inference time. We compare the recognition results with several advanced detection methods: Faster-RCNN (Girshick, 2015), SSD (Fu et al., 2017), and YOLOV5. The experimental results are shown in Table 2.

Swin transformer introduces a hierarchical local attention mechanism, dividing the input sequence into multiple chunks, with global self-attention computation within each chunk and local self-attention computation between different chunks, which can effectively reduce the computational complexity and memory consumption and improve the model efficiency; it also adopts adjustable path length, and by introducing multiple cross-domain feature blocks, feature maps of different sizes can be constructed on different paths, thus improving the expressiveness of the model without increasing the computational complexity; a random Token masking strategy is also introduced to randomly mask a portion of

Table 2 Comparison of STFNet-A with detection methods

Model	Backbone	mAP(%)	FPS	Param(M)
Faster R-CNN	ResNet101	90.5	7.8	189.8
SSD	VGGNet16	81.2	17.5	98.7
YOLOV5	CSPDarkNet53	92.6	41.4	46.9
STFNet-A(Our)	Swin Transformer	96.9	59.6	31.3

Param(M) represents the number of model parameters in millions. FPS is measured with a batch size of 1 under an input resolution of 640×640 after 10 warm-up iterations

Tokens in the input sequence, which can help the model to learn contextual information and semantic relationships better. Since objects may appear in image regions of different scales and proportions in the target detection task, The integration of feature maps from varying scales necessitates a dedicated processing step. The Faster R-CNN relies only on the feature maps of the final layer to achieve target prediction. In contrast, the SSD achieves target detection by employing the multi-layered structure of convolutional neural networks but does not reasonably use the underlying features. In contrast, YOLOV5 adopts the idea of a feature pyramid by stitching together feature maps of different scales and increasing the number of channels to achieve the prediction of targets at three scales. This approach effectively utilizes the underlying features to improve target detection accuracy significantly. The Neck module in this paper uses the feature extractor's last three layers of feature maps to reasonably construct a feature pyramid to fully obtain the category information and global contextual information of the target and extract the local detail information of the object. The prediction module adopts a parameter-sharing design scheme to share the classification and regression parameters at all levels. The introduced CBAM attention module first uses a channel attention mechanism to adjust the importance of each channel in the channel feature map to facilitate the model to capture global information better. It later uses spatial attention to adjust the importance of each pixel in the feature map to facilitate the model to capture spatial information better. This design scheme significantly improves the accuracy of model detection by 96.9%.

Comparing the Faster-RCNN, YOLOV5 and SSD models, the STFNet-A model in this paper performs the best in detection accuracy, with mAP reaching 96.9%. STFNet-A improves by 6.4%, 4.3% and 15.7% compared with these three models. In terms of detection speed, STFNet is also excellent, able to reach 59.6 FPS, which is about 7.6, 1.4, and 3.4 times faster than these three models, respectively; meanwhile, in terms of the number of parameters, STFNet's model parameters are only 31.3M, which is 83.5%, 33.3%, and 15.7% lower than these three models, respectively. 83.5%, 33.3%, and 68.3%, respectively, which means the model constructed in this paper can meet the needs of typical unsafe behavior identification of coal mine workers.

The STFNet-A network achieves a detection speed of 59.6 FPS due to several key factors. It utilizes the Swin Transformer as the backbone, incorporating a hierarchical local attention mechanism that reduces computational complexity and memory usage. The model features adjustable path lengths and multiple cross-domain feature blocks to enhance expressiveness without increased complexity. Additionally, a random token masking strategy aids in learning contextual relationships, while a feature pyramid constructed from the last three feature maps captures both global context and local details. The prediction module employs a parameter-sharing design, and the CBAM attention module optimizes channel and spatial feature importance, significantly boosting detection speed.

3.1.8.2 Visualization Experiments To further validate the reliability and effectiveness of the STFNet-A network model, the recognition results are visualized and analyzed in comparison with target detection models such as YOLOV5, Faster-RCNN, and SSD. This process helps

to evaluate the performance of different models more intuitively and verify the superiority of the STFNet-A network model. The visualization results are shown in Fig. 4, 5, 6, and 7.

From the visualization results, it can be seen that SSD has missed detection in the tiny target and complex background cases, and both err-hat and err-mask categories are not identified. Besides, YOLOV5, Faster-RCNN and SSD did not detect the err-mask category, and all identified the mask category; YOLOV5 identified part of the err-hat category, while Faster-RCNN did not identify the err-hat category. Our STFNet-A is all identified correctly. The visualization results fully demonstrate that the STFNet-A target detection model exhibits good interference suppression capability in complex environments, and this advantage enables it to more effectively meet the challenges in target detection tasks in complex scenarios in coal mines and can meet production requirements.

3.1.8.3 Ablation Experiments In order to further evaluate the feasibility and reliability of the method proposed in this paper, especially to evaluate the performance of each module for constructing the lightweight model STFNet-A, ablation experiments are designed for different modules in this paper, and the experimental results are shown in table Table 3.

After model 1 replaces the backbone network, the detection accuracy and detection speed are decreased, and the number of parameters is increased, which indicates that Swin Transformer is effective as a backbone network; model 2 embeds the attention mechanism before the detection network and the mAP is improved by 1.1% compared with model 1, which indicates that embedding CBAM attention can obtain performance gains; models 3, 4, and 5 at the exact location embedding different attentions, among which embedding CBAM attention is the best, which proves that CBAM retains the most critical information in the



Fig. 4 STFNet-A visualization result



Fig. 5 YOLOV5 visualization result



Fig. 6 Faster R-CNN visualization result



Fig. 7 SSD visualization result

Table 3 Ablation study of STFNet-A components

Model	Method	mAP(%)	FPS	Param(M)
1	STFNet+CSPDarkNet53(Backbone)	91.6	44.9	48.7
2	STFNet+CSPDarkNet53(Backbone)+CBAM	92.7	45.6	48.7
3	STFNet+CBAM	93.8	55.6	31.3
4	STFNet+ECA	92.2	55.1	31.3
5	STFNet+SE	91.4	54.2	31.3
6	STFNet+CBAM+FL	94.1	59.6	31.3
7	STFNet+QFL	95.5	59.6	31.3
8	STFNet+CBAM+QFL	96.9	59.6	31.3

Param(M) represents the number of model parameters in millions. FPS is measured with a batch size of 1 under an input resolution of 640×640 after 10 warm-up iterations

image and suppresses unnecessary information. Models 6, 7, and 8 compare the improved loss function and the embedded CBAM attention. It is found that the QFL loss function can not only consider the quality of samples and assign lower quality samples to greater weights but also take into account the differences between sample classes, and by using dynamic balancing factors, it can balance the distribution of samples within classes and solve the problem of difficult to classify samples. CBAM is adaptive and can automatically learn the important information in images and utilize them. The CBAM can learn important features in images adaptively at different levels and between different channels through the attention mechanism. When this QFL loss and CBAM attention are combined, the best results are achieved, showing that the method in this paper is feasible and effective.

3.2 Alpha-SDL-Pose for Posture-Related Unsafe Behaviors

3.2.1 Alpha-SDL-Pose Overview

In the original AlphaPose, Faster R-CNN is used for human target detection. After our comparative experimental analysis in subsection A, Faster R-CNN needs to be increased in terms of detection speed and the number of parameters to meet the demand for lightweight and real-time detection. In addition, the critical human pose timing information in complex environments (e.g., multiple people, occlusion, low light, etc.) may need to be adequately obtained by the information of feature maps alone. To address these issues, we propose an AlphaPose-based miner posture-related unsafe behaviors recognition network Alpha-SDL-Pose (Alpha_STFNet-A_DNLA_LSTM_Pose), whose structure is shown in Fig. 8.

The proposed Alpha-SDL-Pose aims to extract reliable human pose representations, including keypoint locations, joint relationships, skeletal configurations, and temporal motion information, for posture-related unsafe behaviour recognition. The framework contains three sequential stages: human detection, pose estimation, and unsafe behaviour classification.

In the first stage, STFNet-A is adopted as the human detector to replace the original detector in AlphaPose, providing accurate and efficient localization of miner targets. Based on the detected human regions, the pose estimation module composed of STN (Spatial Transformer Network), SPPE (Single Person Pose Estimator), DNLA (Dual Non-Local Attention), and SDTN (Symmetric Deformable Transformation Network) is employed to extract discriminative pose features. Specifically, the DNLA module enhances the representation of key pose information by modeling spatial dependencies and channel-wise feature responses. After pose estimation, Pose NMS is utilized to select reliable pose predictions. Finally, an LSTM-based classification network is introduced to model temporal pose variations and identify miners' posture-related unsafe behaviours.

LSTM was selected for classifying unsafe posture behaviors among miners due to its ability to effectively handle time series data with long-term dependencies. The dynamic nature of posture changes over time necessitates a model that can capture these temporal characteristics, making LSTM particularly suitable for this task. In contrast, other neural networks may struggle with time-dependent data, and SoftMax classifiers are generally better suited for independent, non-time-series classification tasks. The alert generation mechanism depicted in Fig. 8 is based on the model's predictions across consecutive frames. Alerts are triggered when unsafe posture behaviors are detected for a specified duration or exceed a

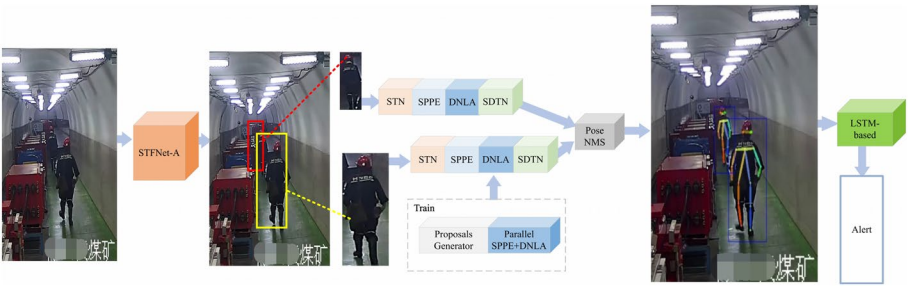


Fig. 8 Alpha-SDL-Pose network structure

defined severity threshold. These thresholds and duration settings are determined according to established safety standards, ensuring the alerts are both accurate and timely, effectively prompting relevant personnel to take necessary preventive measures.

3.2.2 Replacement of Human Detector

In the Alpha-SDL-Pose method proposed in this paper, the human target detection uses our designed lightweight target detection network STFNet-A, which can quickly and accurately extract the human detection frame with a detection speed of 49.4 FPS, which can meet the business requirements.

3.2.3 The Dual Non-Local Attention (DNLA)

Underground coal mines are an extremely complex environment, and performing target feature extraction and enhancing key information is challenging. Images under different lighting conditions may produce similar objects, and the miner targets may appear in the images at different scales when their features, such as colour and brightness, change. The above situation can cause the original Alphapose's pose estimation module to be unable to sufficiently mine to extract important discriminative information, such as the target figure's joint point location, joint connectivity, and skeletal pose. To address the problem of enhancing features with discriminative power to capture longer distance and richer contextual information of the miner's pose action class while suppressing redundant information, this paper designs a Dual Non-Local Attention (DNLA) module, whose structure is shown in Fig. 9.

The input feature map is represented as $F \in \mathbb{R}^{H \times W \times C}$, where H, W, and C denote the height, width, and channel number of the feature map, respectively. As shown in Fig. 9, the feature map is reshaped into two matrix representations, namely $HW \times C$ and $C \times HW$, by flattening the spatial dimensions while preserving the channel dimension. These representations are used for the subsequent attention computations in the spatial and channel branches of the proposed DNLA module.

Compared with the conventional Non-Local Attention (NLA) (Wang et al., 2018), which mainly models spatial dependencies, DNLA introduces an additional channel attention branch to capture channel-wise correlations and enhance discriminative pose features. By jointly modeling spatial and channel relationships, DNLA can better capture the correlation between different locations and feature channels in complex mining scenarios, thereby reducing the influence of redundant information and improving feature representation robustness. Meanwhile, DNLA adaptively weights features from different dimensions, allowing the network to better model the relationships among miner skeletal points and improve the representation of key pose information. For temporal pose modeling, DNLA captures long-range dependencies among pose features by calculating relationships between different positions, resulting in more integrated feature representations for sequential action recognition.

Different from existing dual-attention mechanisms such as CBAM, which mainly recalibrate convolutional feature maps through sequential channel and spatial attention, DNLA is designed specifically for pose estimation under complex mining environments. The spatial branch focuses on modeling long-range relationships among human joints, while the channel branch enhances semantic feature selection by emphasizing informative pose repre-

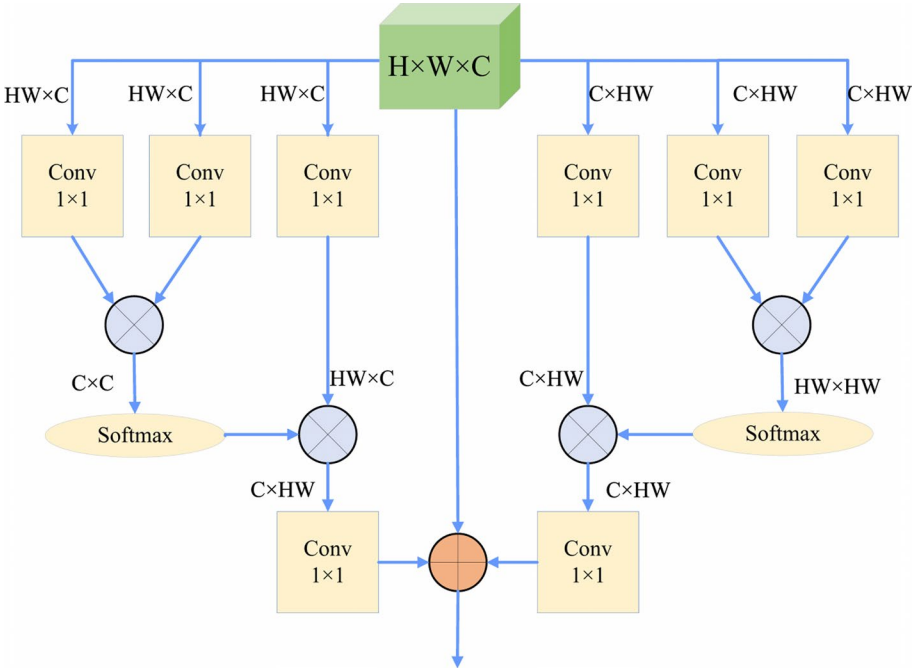


Fig. 9 Structure of the proposed DNLA module. *H*, *W*, and *C* denote the height, width, and channel number of the feature map, respectively

sentations. This design allows the network to maintain discriminative skeletal information when some body parts are occluded or incompletely observed.

Although DNLA introduces an additional channel attention branch, the additional computational cost is expected to be limited because the channel attention operation is performed on the intermediate feature representation rather than the original input image.

The Dual Non-Local Attention module models the relationships between different spatial locations and feature channels to mine discriminative features, such as miners' key joint point locations, joint linkages, and skeletal postures. The attention calculation process is formulated as follows:

$$F_{out} = conv\left(softmax\left(\frac{Q_1K_1}{\sqrt{d_{k1}}}\right)V_1\right) + conv\left(softmax\left(\frac{Q_2K_2}{\sqrt{d_{k2}}}\right)V_2\right) \tag{6}$$

The first term in Eq. (6) corresponds to the spatial attention branch, where the query-key similarity between different spatial positions is calculated to capture long-range dependencies among human joints. This enables the network to model relationships between distant body parts and aggregate contextual information from visible regions. The second term corresponds to the channel attention branch, which models channel-wise correlations and adaptively emphasizes informative feature channels to enhance discriminative pose representations. Finally, the outputs of the spatial and channel attention branches are fused through convolution operations to generate the final feature representation.

Compared with the original Non-Local block, which mainly focuses on spatial dependency modeling, DNLA jointly considers spatial dependencies and channel-wise feature responses. By combining these two complementary attention mechanisms, DNLA can preserve more discriminative skeletal information when human keypoints are partially occluded by mining equipment, protective clothing, or other environmental factors, making it more suitable for posture-related unsafe behaviour recognition in underground mining scenarios.

where Q, K, and V denote the query, key, and value feature representations, respectively. The subscripts 1 and 2 represent the spatial attention branch and channel attention branch, respectively. d_{k1} and d_{k2} denote the dimensions of the key representations in the spatial and channel attention branches, respectively, and F_{out} represents the final feature map obtained after fusing the spatial and channel attention outputs.

3.2.4 LSTM-Based Unsafe Behavior Classification Network

Alpha-SD-Pose is used to obtain the miner target critical time-series pose information, after which long-term time-series features are extracted from the miner target critical time-series pose information using an LSTM-based unsafe behavior classifier to obtain the miner's unsafe behavior. The LSTM-based abnormal behavior classification network technique has significant advantages compared to traditional decision trees and deep forest algorithms. It can extract time-series features with persistence from extensive input data. Its structure consists of two layers with independent parameters in each layer, which makes its overall network parameters better than traditional abnormal behavior classification methods based on skeletal point and joint information, such as graph convolutional neural networks, which are much smaller. In practical applications, the algorithm first maps the input feature information at time t into the fully connected layer feature space, then uses an activation function to nonlinearly transform this part of the feature information, and feeds the obtained feature set into the LSTM unit, and the feature vector generated by the last LSTM unit is also fed into the fully connected layer, followed by Softmax for nonlinear transformation, and finally by One-hot coding is used to obtain the accurate classification of unsafe behaviors.

3.2.5 Datasets

The MSCOCO 2017 (Lin et al., 2014) human pose estimation dataset provides about 200,000 images, over 250,000 human pose annotations, and 1.5 million target instances. The dataset can be used to train and evaluate human pose estimation algorithms. Each human body in the dataset is annotated with joint locations and visibility. There are 17 joints, including left/right shoulder, head, left/right elbow, neck, left/right wrist, left/right ear, left/right hip, left/right knee, left/right eye, nose, and left/right ankle. Visibility is coded as 0 or 1 in the annotation, indicating whether the joint is visible. To better train and evaluate the model, this paper first uses the MSCOCO 2017 human posture estimation dataset to evaluate our model and validate the reliability of the posture evaluation method proposed in this paper, and then visualizes and validates it on the miner posture-related unsafe behaviors dataset. The miner posture-related unsafe behaviors dataset contains 8600 human posture annotations, and the training and testing sets are divided according to 4:1.

3.2.6 Experiment

3.2.6.1 Evaluation metrics The critical human posture assessment is evaluated using the official MSCOCO given target object keypoint similarity (OKS), which is calculated as shown in Eq. (7).

$$OKS = \frac{\sum_i \exp\left(-\frac{d_i^2}{2k_i^2 s^2}\right)\delta(v_i > 0)}{\sum_i \delta(v_i > 0)} \tag{7}$$

Where d_i is the Euclidean distance between the predicted and ground-truth locations of the ith keypoint, v_i represents the visibility flag of the corresponding keypoint, k_i is the keypoint-specific scale constant, s denotes the object scale, and $\delta(v_i > 0)$ is the indicator function.

This paper used mAP as the final evaluation index of miners' posture-related unsafe behaviors, and different AP values are calculated by setting different OKS thresholds where AP@0.5 and AP@0.75 denote the prediction accuracy of the model when OKS is greater than or equal to 0.5 and greater than or equal to 0.75, respectively.

3.2.6.2 Performance Comparison of Different Methods This section aims to compare the performance of the improved human pose estimation model Alpha-SD-Pose with the mainstream human pose estimation model in terms of accuracy and speed. The comparison is based on the MSCOCO 2017 test set. The participating benchmark models are OpenPose (Cao et al., 2017), AlphaPose (Fang et al., 2017), SimpleBaseline-101 (Xiao et al., 2018), and HRNet-W48 (Sun et al., 2019), and the detection speed is tested on the same experimental platform, and the experimental results are shown in Table 4. The results show that the improved Alpha-SD-Pose still has certain advantages in terms of accuracy, and the speed is also substantially improved. In this paper, an improved Alpha-SD-Pose pose estimation method is proposed, which incorporates the lightweight target detection network STFNet-A, thus making the network locate human targets faster; in addition, the Dual Non-Local attention (DNLA) is designed to enhance the key information, thus making the pose estimation more efficient. Our research focuses on identifying miners' posture-related unsafe behaviors, which requires acquiring critical temporal pose information on target human joint point locations, joint linkages, and skeletal postures. In order to obtain this informa-

Table 4 Comparison of pose estimation methods on COCO 2017

Model	AP@0.5(%)	AP@0.75(%)	mAP(%)	FPS	Param(M)
OpenPose	83.2	66.8	61.5	18.6	120.4
AlphaPose	88.7	78.6	72.1	35.8	63.8
Simple-Base-line-101	87.3	76.3	70.5	35.2	57.2
HRNet-w48	89.7	79.4	75.2	30.9	66.7
Alpha-SD-Pose(Our)	88.9	78.9	74.8	49.4	43.9

Param(M) represents the number of model parameters in millions. FPS is measured with a batch size of 1 under an input resolution of 640×640 after 10 warm-up iterations. The reported FPS includes human detection and pose estimation stages

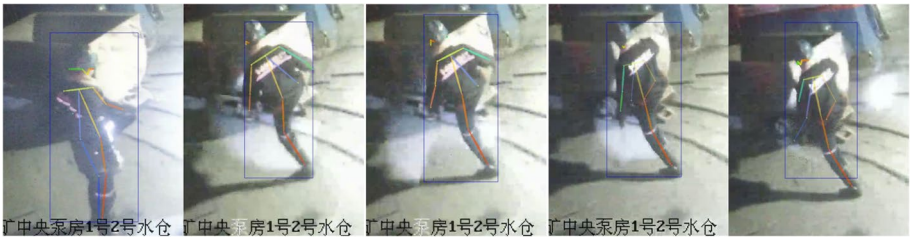


Fig. 10 Visualization result of illegal kicking device



Fig. 11 Visualization results of illegal overturning

tion quickly and accurately, an improved Alpha-SD-Pose network is proposed, which has only 43.6M parameters but achieves a detection accuracy of 74.8%. Although the detection accuracy of Alpha-SD-Pose is slightly lower than that of HRNet-W48 by 0.4%, it has a maximum advantage in computational speed, with a detection speed of 18.5 FPS faster than that of HRNet-W48, reaching 49.4 FPS. In summary, the Alpha-SD-Pose network performs well in miner gesture-like unsafe behavior recognition, with high detection accuracy and computational speed, and the number of parameters is small. Although it is slightly deficient in some aspects compared with some mainstream models, the proposed Alpha-SD-Pose model in this paper is competitive when accuracy and computational speed are considered together.

In order to verify the accuracy of miner posture-related unsafe behaviors recognition, this paper uses the constructed miner posture-related unsafe behavior recognition network Alpha-SDL-Pose to visualize the miner posture-related dataset, and the specific results are shown in Figs. 10 and 11.

The experimental results, as well as the visualization results, show that Alpha-SDL-Pose can meet this part of the business requirements, and Alpha-SDL-Pose incorporated into STFNet-A for lightweight target detection can significantly improve the speed of detecting human bodies and require less computational and storage resources to improve the efficiency and performance of the model. The Dual Non-Local attention (DNLA) designed in this paper not only takes into account location attention but also increases the use of channel attention. Channel and position attention can better express the correlation between different positions and channels in the input miner's temporal pose information sequence. DNLA can globally capture the relevant information in images and videos to understand better and model the whole scene and better handle long-range dependencies in better performance in

Table 5 Ablation study of DNLA and STFNet-A components

Model	Method	mAP(%)	FPS	Param(M)
1	AlphaPose	89.8	35.8	63.8
2	AlphaPose+STFNet-A	92.3	49.4	42.5
3	AlphaPose+STFNet-A+DNLA	94.9	49.4	43.9

Param(M) represents the number of model parameters in millions. FPS is measured with a batch size of 1 under an input resolution of 640×640 after 10 warm-up iterations, including both human detection and pose estimation stages

handling critical pose timing information of miners. With the above strategies, the DNLA attention module can achieve channel attention and position attention to complement each other and have more substantial suppression of noise and interference in the input to improve the expressiveness, robustness, generalization performance and accuracy of the model.

3.2.6.3 Ablation Experiments To test the performance of the pose behavior estimation algorithm Alpha-SDL-Pose, which incorporates STFNet-A and DNLA, such as accuracy and detection speed, ablation experiments are conducted on the miner posture-related unsafe behavior dataset, and the experimental results are shown in Table 5. Model 1 represents the original AlphaPose, model 2 represents human target detection using the lightweight target detection algorithm STFNet-A constructed in this paper, replacing the original Faster R-CNN, and model 3 represents a fusion of the lightweight target detection network STFNet-A and a Dual Non-Local attention pose estimation network. All the above three methods finally use the LSTM-based unsafe behavior classification network to output the target task-specific dangerous actions. The Alpha-SDL-Pose proposed in this paper has apparent advantages in terms of detection speed, accuracy and parameters for two reasons. On the one hand, because the original Faster R-CNN two-stage detection algorithm has the natural disadvantage of slow detection speed and a large number of AlphaPose framework parameters, the STFNet-A target detection network balances speed and accuracy; on the other hand, the Dual Non-Local attention module enhances the pose representation by modeling spatial dependencies and channel-wise feature responses, thus improving the recognition accuracy of posture-related unsafe behaviors.

The improvement from model 2 to model 3 demonstrates the contribution of the DNLA module. By introducing the channel attention branch into the original Non-Local mechanism, DNLA provides complementary feature representation beyond spatial dependency modeling. The channel branch enhances the selection of discriminative pose-related features, while the spatial branch captures long-range dependencies among human joints. Therefore, DNLA can improve the robustness of pose feature extraction when keypoint information is incomplete or affected by occlusion in underground mining environments.

4 Discussion

In this study, we propose an Unsafe Behaviors Monitoring Network (UBM-Net), which integrates the object detection network STFNet-A and the human pose estimation network Alpha-SD-Pose. The network is designed to detect miners and identify their unsafe behaviors to enhance safety in coal mining operations. The proposed framework combines

appearance-based detection and skeleton-based pose estimation because these two types of information provide complementary cues for unsafe behavior recognition. Skeleton-based methods are effective in modeling human motion when body keypoints can be accurately localized. However, underground mining environments often involve low illumination, severe occlusion caused by mining equipment, and interference from protective equipment, which may reduce the reliability of keypoint localization. In contrast, appearance information, such as helmets, masks, mining tools, and surrounding scene context, provides additional safety-related cues that cannot be captured by skeleton representations alone. Therefore, integrating appearance and pose information enables UBM-Net to achieve more robust unsafe behavior recognition in complex mining environments.

Nevertheless, we acknowledge that the subjectivity in defining unsafe behaviors may influence the results. Different experts or regions may have varying perspectives on the risk levels associated with certain behaviors. To address this issue, future research will incorporate a broader range of expert opinions, establish more standardized criteria for unsafe behavior classification, and further expand the miner unsafe behavior dataset to cover a wider range of hazardous scenarios.

Although STFNet-A and Alpha-SDL-Pose were evaluated separately to verify the effectiveness of each module, MIUBM-Net is designed as an integrated framework for comprehensive unsafe behavior monitoring. Due to the scope of the current study, a complete end-to-end evaluation of the integrated framework on continuous surveillance videos was not conducted. Future work will further investigate the interaction between the two modules and evaluate the overall performance of MIUBM-Net in practical deployment scenarios.

Although the proposed method demonstrates promising performance on the constructed miner unsafe behavior dataset, its generalization to other public datasets has not been extensively evaluated in this work. Since the proposed framework is specifically designed for underground mining scenarios, future work will further validate its effectiveness on additional public benchmarks and cross-domain datasets to investigate its generalization capability.

5 Conclusion

Identifying miners' unsafe behaviors is a critical task in the complex working scenarios of coal mines. This study explores approaches to identify miners' unsafe behaviors by leveraging video surveillance systems deployed in coal mines. The research in this paper develops a miner unsafe behavior identification dataset, classifying behaviors into typical unsafe behaviors and posture-related unsafe behaviors based on coal mine safety risk standards. Because of the difficulty of current detection methods to detect miners' typical unsafe behaviors accurately and in real-time in the complex operating environment of coal mines, an effective target detection model STFNet-A is proposed, using Swin Transformer as the backbone network and constructing a feature pyramid, which not only introduces Quality Focal Loss loss function to focus on the intra-class differences of samples. The accuracy of conventional unsafe behavior recognition reaches 96.9%, the detection speed is 59.6 FPS, and the number of model parameters is 31.3M. To address the challenge of posture-based unsafe behavior recognition among miners, we propose an optimized model based on AlphaPose, named Alpha-SDL-Pose. Firstly, the human detector Faster R-CNN of AlphaPose

is replaced by STFNet-A to improve the detection speed and reduce the number of model parameters. Then, we design the Dual Non-Local Attention (DNLA) module and embed the pose estimator to extract the feature information such as the target person's joint point location, joint linkage and skeletal pose. Finally, an LSTM-based unsafe behavior classification network is used to classify dangerous actions accurately. The model improves the recognition accuracy by 2.7 percentage points over AlphaPose to 74.8%, and increases the detection speed from OpenPose's 18.6 FPS to 49.4 FPS on the MSCOCO 2017 public dataset of over 250,000 human pose samples. The recognition accuracy of miners' posture-like unsafe behaviors on the homemade dataset reaches 94.9%, and the model parameters are 43.9M.

Experimental results demonstrate the effectiveness of STFNet-A and Alpha-SDL-Pose in recognizing typical and posture-related unsafe behaviors, respectively. By integrating these two complementary modules, MIUBM-Net provides a practical framework for intelligent unsafe behavior monitoring in underground mining environments. A comprehensive end-to-end evaluation of the integrated framework on continuous surveillance videos will be investigated in future work.

Acknowledgements This work was supported by High Performance Computing Platform of Henan University of Technology.

Author Contributions W.L., H.X., Z.L., and S.W. wrote the main manuscript text. W.L. and S.W. prepared the methodology and software. H.X. and Z.L. performed validation and investigation. D.Z. supervised the study and curated the data. All authors reviewed the manuscript.

Data Availability The data used in this study consists of a self-constructed mining safety behavior dataset, which contains anonymized on-site surveillance video frames collected from real mining environments, augmented with synthetic samples. Due to privacy and safety regulations, the dataset cannot be made publicly available. However, we provide a detailed description of the data collection, annotation process, and preprocessing pipeline in the manuscript to ensure reproducibility of our work.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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